ARJUNA JEE 2027

Lecture -09 | Mathematics

SEQUENCE & SERIES

By- Sachin Jakhar Sir



Topics *to be covered*

A Miscellaneous Series

B Method of Subtraction

C Question Practice

D



Question Practice

Question

• Multiple layer AGP

If $|x| < 1$, then compute S_∞ :

$$S_\infty = 1^2 + 2^2x + 3^2x^2 + 4^2x^3 + \ldots\ldots \infty = \frac{1+x}{(1-x)^3}$$ ✓

$$xS_\infty = 0 + 1^2x + 2^2x^2 + 3^2x^3 + \ldots\ldots \infty$$

$$\ominus$$

$$S_\infty - xS_\infty = 1^2 + (2^2-1^2)x + (3^2-2^2)x^2 + (4^2-3^2)x^3 + \ldots\ldots \infty$$

$$S_\infty(1-x) = 1 + 3x + 5x^2 + 7x^3 + 9x^4 + \ldots \infty$$

$$S_\infty(1-x) = 1 + 3x + 5x^2 + 7x^3 + 9x^4 + \ldots \infty = \frac{1+x}{(1-x)^2}$$

$$\# \quad M = 1 + 3x + 5x^2 + 7x^3 + 9x^4 + \ldots \infty$$

$$\ominus \quad Mx = 0 + x + 3x^2 + 5x^3 + 7x^4 + \ldots \infty$$

$$M(1-x) = 1 + 2x + 2x^2 + 2x^3 + 2x^4 + \ldots \infty$$

$$= 1 + 2\left(x + x^2 + x^3 + \ldots \infty\right)$$

$$M(1-x) = 1 + 2\frac{x}{1-x} = \frac{1-x+2x}{1-x} = \frac{1+x}{1-x} \Rightarrow M = \frac{1+x}{(1-x)^2}$$

Question [JEE Mains-2014]

80%

If $(10)^9 + 2(11)^1(10)^8 + 3(11)^2(10)^7 + ... + 10(11)^9 = k(10)^9$, then k =

A. 100

B. 110

C. $\frac{121}{10}$

D. $\frac{441}{100}$

$S = (10)^9 + 2(11)^1 \cdot 10^8 + 3\,11^2 \cdot 10^7 + \ldots\ldots\ldots + 9 \cdot 10^1 \cdot 11^8 + 10 \cdot 11^9$

$\times \frac{11}{10}$

$\frac{11S}{10} = 0 + 11^1 \cdot 10^8 + 2 \cdot 11^2 \cdot 10^7 + \ldots\ldots\ldots + 9 \cdot 11^9 + 11^{10}$

$S - \frac{11S}{10} = \ldots\ldots$

Question [JEE Mains-2023]

Multiple layer A.G.P. + Calculative

For $k \in \mathbb{N}$, if the sum of the series $1 + \frac{4}{k} + \frac{8}{k^2} + \frac{13}{k^3} + \frac{19}{k^4} + \cdots$ is 10, then the value of k is

$S = 10$.

$$S = 1 + \frac{4}{k} + \frac{8}{k^2} + \frac{13}{k^3} + \frac{19}{k^4} + \ldots\ldots\infty$$

$\frac{1}{k} \times$

$$\frac{S}{k} = 0 + \frac{1}{k} + \frac{4}{k^2} + \frac{8}{k^3} + \frac{13}{k^4} + \ldots\ldots\infty$$

$\ominus$

$$S - \frac{S}{k} = 1 + \frac{3}{k} + \frac{4}{k^2} + \frac{5}{k^3} + \frac{6}{k^4} + \ldots\infty$$

81%

$$\xi - \frac{\xi}{K} = 1 + \frac{3}{K} + \frac{4}{K^2} + \frac{5}{K^3} + \frac{6}{K^4} + \ldots \infty$$

$$M = \frac{3}{K} + \frac{4}{K^2} + \frac{5}{K^3} + \frac{6}{K^4} + \ldots \infty$$

$\frac{1}{K} \times$

$$\frac{M}{K} = 0 + \frac{3}{K^2} + \frac{4}{K^3} + \frac{5}{K^4} + \ldots \infty$$

$\ominus$

$$M - \frac{M}{K} = \frac{3}{K} + \boxed{\frac{1}{K^2} + \frac{1}{K^3} + \frac{1}{K^4} + \ldots \infty}$$

$$M\left(1 - \frac{1}{K}\right) = \frac{3}{K} + \left(\frac{\frac{1}{K^2}}{1 - \frac{1}{K}}\right) \Rightarrow M\left(\frac{K-1}{K}\right) = \frac{3}{K} + \frac{1}{K(K-1)} \Rightarrow M = \frac{3}{K-1} + \frac{1}{(K-1)^2}$$

$$S - \frac{S}{K} = 1 + \frac{3}{K} + \frac{4}{K^2} + \frac{5}{K^3} + \frac{6}{K^4} + \ldots \infty$$

$$S\left(\frac{K-1}{K}\right) = 1 + \frac{3}{K-1} + \frac{1}{(K-1)^2}$$

$$S = \frac{K}{K-1} + \frac{3K}{(K-1)^2} + \frac{K}{(K-1)^3} = 10, \quad K \in N$$

$\Rightarrow K = 2$

$RHS = 2 + 3\times 2 + 2 = 10$

$K = 2$ fnf

{ Take LCM ⇓ form cubic ⇓ APPLY RRT. }



LAST CLASS RECAP

Answer the following Questions:

1. $\sum_{r=1}^{10}(3r^2-5r+2)=3\sum_{r=1}^{10}r^2-5\sum_{r=1}^{10}r+\sum_{r=1}^{10}2$ → (T)

2. $\sum_{r=1}^{n}r(2r+3)=(\sum_{r=1}^{n}r)\cdot(\sum_{r=1}^{n}2r+3)$ → (F)

3. $\sum_{r=1}^{n}7=7n$ → (T)

4. $\sum_{r=0}^{n}7=7(n+1)$ → (T)



LAST CLASS RECAP

Important Results to Remember as Results (RAR):

A. $\sum_{r=1}^{n} r = \underset{\uparrow}{1} + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$

B. $\sum_{r=1}^{n} r^2 = \underset{\uparrow}{1^2} + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$

C. $\sum_{r=1}^{n} r^3 = \underset{\uparrow}{1^3} + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$

IMPORTANT RESULTS

$$\sum_{r=1}^{n}(2r-1) = 1+3+5+\ldots+(2n-1) = n^2$$

A.P. $a=1, d=2, n$

= Sum of first n-odd Natural Numbers

$$\sum_{r=1}^{n} 2r = 2+4+6+\ldots.+(2n) = (n(n+1))$$

$a=2, d=2, n$

= Sum of first n-even Natural Numbers

$$\sum_{r=1}^{n} a^r = a^1+a^2+a^3+\ldots.+a^n = a\left(\frac{a^n-1}{a-1}\right)$$

G.P.

= Sum of G.P.

RAPID FIRE ROUND

1. $\sum_{r=1}^{15} r^2 = \sum_{n=1}^{15} n^2 = 1^2 + 2^2 + 3^2 + \ldots + 15^2 = \frac{15(15+1)(2\times15+1)}{6}$ fuf

$n = 15$

2. $\sum_{r=9}^{15} r^2 =$

$n.o.t. = 7$

$\sum_{n=9}^{15} n^2 = 9^2 + 10^2 + 11^2 + \ldots + 15^2 = \frac{7(7+1)(2\times7+1)}{6}$ ✗ Wrong

$\left(1^2+2^2+3^2+\ldots+8^2+9^2+10^2+11^2+\ldots+15^2\right) - \left(1^2+2^2+3^2+\ldots+8^2\right)$

$n = 15$ $\qquad$ $n = 8$

$\left(\frac{15\times(15+1)(2\times15+1)}{6} - \frac{8(8+1)(2\times8+1)}{6}\right)$ fnf

RAPID FIRE ROUND

3. $\text{Sum}(S) = 2^3 + 4^3 + 6^3 + 8^3 + \cdots + 20^3$

4. $\sum_{r=1}^{10}(2r-1)^3 =$

$$\sum_{r=1}^{10}(2r-1)^3 = 1^3+3^3+5^3\ldots+19^3$$

$2^3 = 1^3 \times 2^3$

$4^3 = 2^3 \times 2^3$

$6^3 = 2^3 \times 3^3$

Sumit:

$$S = \left(1^3+2^3+3^3+4^3+\ldots\ldots+20^3\right) - \left(1^3+3^3+5^3+\ldots+19^3\right)$$

Santosh:

$$S = 2^3\left(1^3+2^3+3^3+4^3+\ldots\ldots+10^3\right)$$

$n=10$

$$S = 8\left(\frac{10\times 11}{2}\right)^2$$

$$\sum_{r=1}^{10}(2r-1)^3 = 1^3+3^3+5^3\ldots+19^3$$

$$= \left(1^3+2^3+3^3+4^3+5^3+6^3+\ldots+18^3+19^3\right) - \left(2^3+4^3+6^3+\ldots+18^3\right)$$

$n=19$

$$\left(\frac{19\times 20}{2}\right)^2 - 8\left(\frac{9\times 10}{2}\right)^2$$

$$-2^3\left(1^3+2^3+3^3+\ldots+9^3\right)$$

$n=9$

Always Method.

$$\sum_{r=1}^{10}(2r-1)^3 = \sum_{r=1}^{10}8r^3 - \sum_{r=1}^{10}1 + \sum_{r=1}^{10}6r - \sum_{r=1}^{10}12r^2$$

Same ✓

$$= 8\left(\sum_{r=1}^{10}r^3\right) - (10) + 6\sum_{r=1}^{10}r - 12\sum_{r=1}^{10}r^2$$

$\left(1^3+2^3+\ldots+10^3\right)$ $\quad 1+2+3+\ldots+10$ $\quad 1^2+2^2+\ldots+10^2$

Question [JEE Mains-2020]

[Ans. : 504]

The sum, $\sum_{n=1}^{7} \frac{n(n+1)(2n+1)}{4}$ is equal to

$$\frac{1\times2\times3}{4} + \frac{2\times3\times5}{4} + \frac{3\times4\times7}{4} + \ldots\ldots \text{ upto } 7 \text{ terms}$$

$$\frac{(n^2+n)(2n+1)}{4}$$

$$\frac{2n^3+2n^2+n^2+n}{4} = \frac{2n^3+3n^2+n}{4}$$

$$\frac{1}{4}\sum_{n=1}^{7}(2n^3+3n^2+n) = \frac{1}{4}\left(2\sum_{n=1}^{7}n^3 + 3\sum_{n=1}^{7}n^2 + \sum_{n=1}^{7}n\right) = \frac{1}{4}\left(2\times\left(\frac{7\times8}{2}\right)^2 + 3\times\frac{7\times8\times15}{6} + \frac{7\times8}{2}\right)$$

$1^3+2^3+\ldots+7^3$ $\quad$ $1^2+2^2+\ldots+7^2$ $\quad$ $1+2+3+\ldots+7$

Question [JEE Mains-2020]

[Ans. : 1540]

The sum $\sum_{k=1}^{20}(1+2+3+\ldots+k)$ is

$$\sum_{k=1}^{20}\frac{k(k+1)}{2} = \frac{1}{2}\left(\sum_{k=1}^{20}k^2 + \sum_{k=1}^{20}k\right)$$

$$= \frac{1}{2}\left(\frac{20\times21\times41}{6} + \frac{20\times21}{2}\right)$$

Acidic Question on Multiple Sigma

Compute the value of: $\sum_{i=1}^{n} \sum_{j=1}^{i} \sum_{k=1}^{j} 1$

$$\sum_{i=1}^{n} \sum_{j=1}^{i} \left(\sum_{k=1}^{j} 1 \right) = \sum_{i=1}^{n} \left(\sum_{j=1}^{i} (j) \right) = \sum_{i=1}^{n} \frac{i(i+1)}{2} = \frac{1}{2} \left(\sum_{i=1}^{n} (i^2 + i) \right)$$

$(1 + 1 + 1 + \ldots \underbrace{1}_{j\text{-Times}})$

$1 + 2 + 3 + \ldots + i = \frac{i(i+1)}{2}$

$$\frac{1}{2} \left(\sum_{i=1}^{n} i^2 + \sum_{i=1}^{n} i \right)$$

$1^2 + 2^2 + \ldots + n^2$ $\qquad$ $1 + 2 + 3 + \ldots n$

$$\frac{1}{2} \left(\frac{n \times (n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \right)$$

Acidic Question on Multiple Sigma

Compute the value of : $S = \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \frac{1}{2^i . 3^j 5^k}$

HW.

$$\sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \sum_{i=0}^{\infty} \left(\frac{1}{2^i} \cdot \frac{1}{3^j} \cdot \frac{1}{5^k} \right)$$

Challenger Question [JEE Mains-2022]

If $A = \sum_{n=1}^{\infty} \frac{1}{(3+(-1)^n)^n}$ and $B = \sum_{n=1}^{\infty} \frac{(-1)^n}{(3+(-1)^n)^n}$, then $\frac{A}{B}$ is equal to :

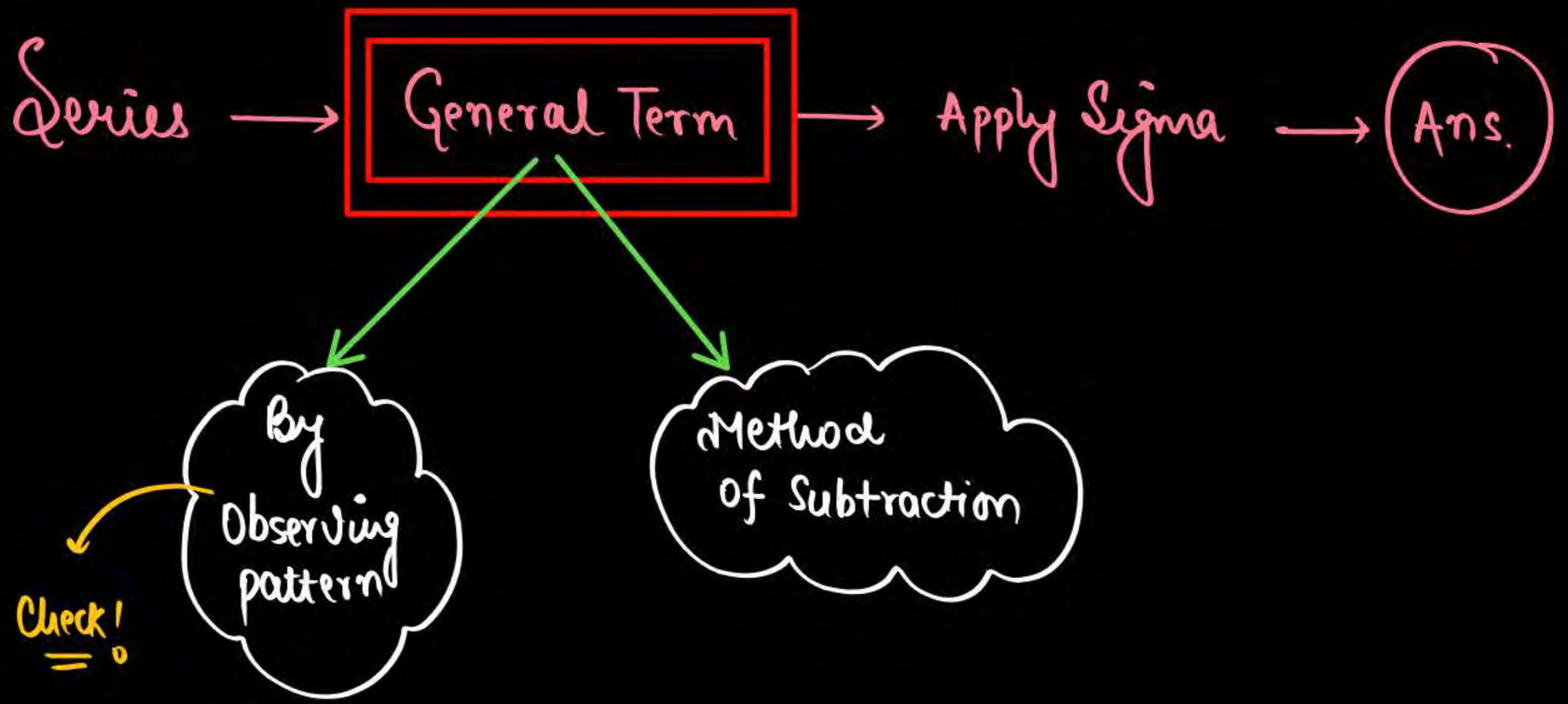
A $\frac{11}{9}$

B 1

C $-\frac{11}{9}$

D $-\frac{11}{3}$

MISCELLAENOUS SEQUENCES



Question 1

Compute $S = \underbrace{1 \times 3}_{T_1} + \underbrace{2 \times 5}_{T_2} + \underbrace{3 \times 7}_{T_3} + \underbrace{4 \times 9}_{T_4} + \ldots\ldots$ upto 18 terms.

$r=1$, $r=2$, $r=3$

$T_n = n(2r+1)$

$$\# \quad S = \sum_{r=1}^{18} n(2r+1) = \sum_{r=1}^{18} (2r^2 + r)$$

$$S = 2\sum_{r=1}^{18} n^2 + \sum_{r=1}^{18} n$$

$1^2 + 2^2 + \ldots + 18^2$

$$S = \left(2 \times \left(\frac{18 \times 19 \times 37}{6}\right) + \frac{18 \times 19}{2}\right) \text{ Ans.}$$

Question 2

Evaluate : $S = \overbrace{1}^{T_1} + \overbrace{(1+2)}^{T_2} + \overbrace{(1+2+3)}^{T_3} + \overbrace{(1+2+3+4)}^{T_4} + \ldots\ldots$ upto 10 terms.

$$T_n = 1+2+3+\ldots\ldots+n = \frac{n(n+1)}{2}$$

$n=1 \quad n=2 \quad n=3$

$T_1 = 1 \quad T_2 = 1+2 \quad T_3 = 1+2+3$

$n=1 \rightarrow \frac{1\times 2}{2} = 1$

$n=2 \rightarrow \frac{2\times 3}{2} = 3$

$n=3 \rightarrow \frac{3\times 4}{2} = 6$

$n=4 \rightarrow \frac{4\times 5}{2} = 10$

$$S = \sum_{r=1}^{10} \frac{n(n+1)}{2} = \frac{1}{2}\left(\sum_{r=1}^{10} n^2 + \sum_{r=1}^{10} n\right)$$

Question [JEE Mains-2019]

The sum $1+\frac{1^3+2^3}{1+2}+\frac{1^3+2^3+3^3}{1+2+3}+\ldots+\frac{1^3+2^3+3^3+\ldots+15^3}{1+2+3+\ldots+15}-\frac{1}{2}(1+2+3+\ldots+15)$ is equal to

(T_1, T_2, T_3, T_{15})

A 1860

B 620

C 660

D 1240

$T_1=\frac{1^3}{1}=1$

$T_2=\frac{1^3+2^3}{1+2}=\frac{1+8}{1+2}=3$

$T_3=\frac{1^3+2^3+3^3}{1+2+3}=\frac{1+8+27}{6}=6$

$T_4=\frac{1^3+2^3+3^3+4^3}{1+2+3+4}=10$

$T_n=\frac{1^3+2^3+3^3+\ldots+n^3}{1+2+3+\ldots+n}=\frac{\left(\frac{n(n+1)}{2}\right)^2}{\frac{n(n+1)}{2}}=\frac{n(n+1)}{2}$

$T_n=\frac{n(n+1)}{2}$

$n=1 \Rightarrow T_1=1$

$n=2 \Rightarrow T_2=3$

$n=3 \Rightarrow T_3=6$

Question [JEE Mains-2019]

The sum $1+\frac{1^3+2^3}{1+2}+\frac{1^3+2^3+3^3}{1+2+3}+\ldots+\frac{1^3+2^3+3^3+\ldots+15^3}{1+2+3+\ldots+15}-\frac{1}{2}(1+2+3+\ldots+15)$ is equal to

A 1860

B 620

C 660

D 1240

$$\sum_{n=1}^{15}\frac{n(n+1)}{2}-\frac{1}{2}\left(\frac{15\times16}{2}\right)$$

$$\frac{1}{2}\left(\sum_{n=1}^{15}n^2+\sum_{n=1}^{15}n\right)-\frac{1}{2}\times\left(\frac{15\times16}{2}\right)$$

Q IBY → (P)

Question [JEE Mains-2019]

Ans: (B)

The sum of the following series

$$1+6+\frac{9(1^2+2^2+3^2)}{7}+\frac{12(1^2+2^2+3^2+4^2)}{9}+\frac{15(1^2+2^2+....+5^2)}{11}+\text{.... up to 15 terms is}$$

T_3 T_4 T_5

A 7830

B 7820

C 7520

D 7510

$$T_n=\frac{3r\left(1^2+2^2+3^2+....+n^2\right)}{(2r+1)}$$

$n=1$

$$T_1=\frac{3(1^2)}{3}\checkmark$$

$r=2 \rightarrow T_2=\frac{6(1^2+2^2)}{5}=6$

$$=\frac{3r\left(n(n+1)(2r+1)\right)}{6\,(2r+1)}$$

$$T_n=\frac{n^3+r^2}{2}$$

$$T_2=\frac{6\times(1^2+2^2)}{5}$$

$$T_1=\frac{3\times(1^2)}{3}$$

Question [JEE Mains-2019]

Ans: (B)

The sum of the following series

$1 + 6 + \frac{9(1^2+2^2+3^2)}{7} + \frac{12(1^2+2^2+3^2+4^2)}{9} + \frac{15(1^2+2^2+....+5^2)}{11}$ + ... up to 15 terms is

(T_3, T_4, T_5)

A. 7830

B. 7820

C. 7520

D. 7510

$$S = \sum_{r=1}^{15} \frac{r^3+r^2}{2} = \frac{1}{2}\left(\sum_{r=1}^{15} r^3 + \sum_{r=1}^{15} r^2\right) \checkmark$$

$$= \frac{1}{2}\left(\left(\frac{15\times16}{2}\right)^2 + \frac{15\times16\times31}{6}\right)$$

Alternate $\oplus$ / $\ominus$ Sign $\rightarrow$ $\boxed{(-1)^r}$

$$\sum_{n=1}^{10} n = 1+2+3+4+\ldots+10.$$

$$\sum_{r=1}^{10} (-1)^r \cdot r = -1+2-3+4-5+6\ldots$$

Question [JEE Mains-2022]

[Ans. : 120]

$\frac{2^3-1^3}{1\times 7}+\frac{4^3-3^3+2^3-1^3}{2\times 11}+\frac{6^3-5^3+4^3-3^3+2^3-1^3}{3\times 15}+\cdots+\frac{30^3-29^3+28^3-27^3+\cdots+2^3-1^3}{15\times 63}$ is equal to

$T_1=\frac{2^3-1^3}{1\times 7}=\frac{8-1}{7}=1$ ✓

$T_2=\frac{4^3-3^3+2^3-1^3}{22}=\frac{44}{22}=2$ ✓

$T_3=\frac{6^3-5^3+4^3-3^3+2^3-1^3}{45}=\frac{135}{45}=3$ ✓

$1+2+3+\ldots\ldots+15$

$=\frac{15\times \overset{8}{\cancel{16}}}{\cancel{2}}=120$

Terms Complicated
↓
Simplify before G.T.

Question [JEE Mains-2025]

Ans: B

$1 + 3 + 5^2 + 7 + 9^2 + \ldots$ upto 40 terms is equal to

A. 43890

B. 41880

C. 33980

D. 40870

$S = 1^2 + 3 + 5^2 + 7 + 9^2 + 11 + 13^2 + \ldots$ upto 40 Terms.

$S = (1^2 + 5^2 + 9^2 + 13^2 + \ldots) + (3 + 7 + 11 + 15 + \ldots)$

$(1^2 + 5^2 + 9^2 + 13^2 + \ldots)$ → (X)

$(3 + 7 + 11 + 15 + \ldots)$ → A.P.

$a = 3$

$d = 4$ ✓

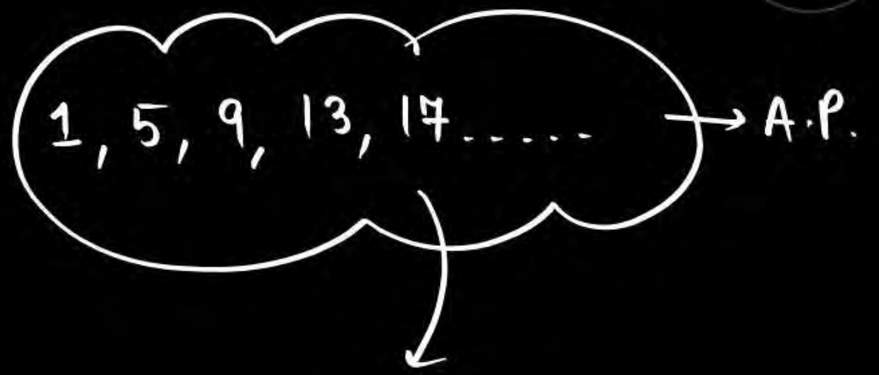
$n = 20$

$$X = (1^2 + 5^2 + 9^2 + 13^2 + \ldots)$$

$$X = \sum_{n=1}^{20} (4n-3)^2$$

$$X = \sum_{n=1}^{20} (16n^2 + 9 - 24n)$$

$$\boxed{X = 16\sum_{n=1}^{20} n^2 + \sum_{n=1}^{20} 9 - 24\sum_{n=1}^{20} n}$$



$$T_n = 1 + (n-1)4$$

$$= 4n - 3$$

Question [JEE Mains-2025]

[Ans. : (C)]

If $\frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+\cdots\infty=\frac{\pi^4}{90}$,

$\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots\infty=\alpha$,

$\frac{1}{2^4}+\frac{1}{4^4}+\frac{1}{6^4}+\cdots\infty=\beta$, then $\frac{\alpha}{\beta}$ is equal to



A 23

B 18

C 15

D 14

DO IT BY YOURSELF (DIBY)

GEOMETRIC PROGRESSION

DIBY-43

If the sum of an infinite GP a, ar, ar^2, ar^3....., is 15 and the sum of squares of of its each term is 150, then the sum of of ar^2, ar^4, ar^6.....upto infinity, is:

[JEE Mains-2021]

(A) $\frac{1}{2}$ (B) $\frac{25}{2}$ (C) $\frac{5}{2}$ (D) $\frac{9}{2}$

[Ans. : (A)]

DIBY-44

Three numbers are in an increasing geometric progression with common ratio r. If the middle number is doubled, then the new numbers are in an arithmetic progression with common difference d. If the fourth term of GP is $3r^2$, then $r^2 - d$ is equal to:

[JEE Mains-2021]

(A) $7 + \sqrt{3}$ (B) $7 - \sqrt{3}$ (C) $7 + 3\sqrt{3}$ (D) $7 - 7\sqrt{3}$

[Ans. : (A)]

DIBY-45

If the sum of the second, third and fourth terms of a positive term. G.P. is 3 and the sum of its sixth, seventh and eighth terms is 243, then the sum of the first 50 terms of this G.P. is: [JEE Mains-2020]

(A) $\frac{2}{13}(3^{50}-1)$ (B) $\frac{1}{13}(3^{50}-1)$

(C) $\frac{1}{26}(3^{49}-1)$ (D) $\frac{1}{26}(3^{50}-1)$ [Ans. : (D)]

DIBY-46

The greatest positive integer k, for which 49^k+1 is a factor of the sum $49^{125}+49^{124}+\cdots+49^2+49+1$, is [JEE Mains-2020]

(A) 32 (B) 63 (C) 65 (D) 60

[Ans. : (B)]

DIBY-47

If the sum of the second, fourth and sixth terms of a G.P. of positive terms is 21 and the sum of its eighth, tenth and twelfth terms is 15309, then the sum of its first nine terms is: [JEE Mains-2025]

(A) 760 (B) 755 (C) 750 (D) 757

[Ans. : (D)]

DIBY-48

If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct? [JEE Mains-2019]

(A) d, e, f are in A.P.

(B) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.

(C) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.

(D) d, e, f are in G.P.

[Ans. : (C)]

DIBY-49

Find the sum of the following series

(i) 5 + 55 + 555 + ... to n terms

(ii) 0.3 + 0.33 + 0.333 + ... to n terms

[Ans. : (i) $\frac{5}{9}\left[10\left(\frac{10^n-1}{9}\right) - n\right]$ (ii) $\frac{1}{3}\left[n - \frac{1}{9}\left(1 - \frac{1}{10^n}\right)\right]$]

DIBY-50

If $x = 1 + a + a^2 + a^3 + \ldots\ldots \infty$ and $y = 1 + b + b^2 + b^3 + \ldots\ldots \infty$ show that

$1 + ab + a^2b^2 + a^3b^3 + \ldots\ldots \infty = \frac{xy}{x+y-1}$ where $0 < a < 1$ and $0 < b < 1$.

DIBY-51

For the two positive number a, b if a, b and $\frac{1}{18}$ are in a geometric progression, while $\frac{1}{a}$, 10 and $\frac{1}{b}$ are in an arithmetic progression, then $16a + 12b$ is equal to

[JEE Mains-2023]

[Ans. : 3]

DIBY-52

If the first 3 consecutive terms of a geometrical progression are the roots of the equation $2x^3 - 19x^2 + 57x - 54 = 0$ find the sum to infinite number of terms of G.P.

[Ans. : $\frac{27}{2}$]

DIBY-53

Let x_1, x_2, x_3, x_4 be in a geometric progression. If $2, 7, 9, 5$ are subtracted respectively from x_1, x_2, x_3, x_4 then the resulting numbers are in an arithmetic progression. Then the value of $\frac{1}{24}(x_1x_2x_3x_4)$ is: [JEE Mains-2025]

(A) 72 (B) 18 (C) 36 (D) 216

[Ans. : 216]

DIBY-54

If p arithmetic mean are inserted between 5 and 41 so that ratio $\frac{A_3}{A_{p-1}} = \frac{2}{5}$, then find the value of p.

[Ans. : 11]

DIBY-55

If $a_1 a_2 a_3 \ldots\ldots a_n$ are n arithmetic means inserted between 7 and 2015 whose sum is 56616, then

(A) n is 56

(B) n is 28

(C) $a_{19} = 2029/3$

(D) $a_{19} = 36543/57$

[Ans. : (A,C)]

DIBY-56

The sides of a right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side?

[Ans. : 6]

A.G.P.

DIBY-57

Let $S = 109 + \frac{108}{5} + \frac{107}{5^2} + \cdots + \frac{2}{5^{107}} + \frac{1}{5^{108}}$. Then the value of $(16S - (25)^{-54})$ is equal to

[JEE Mains-2023]

[Ans. : 2175]

DIBY-58

Suppose $a_1, a_2, 2, a_3, a_4$ be in an arithmetico-geometric progression. If the common ratio of the corresponding geometric progression is 2 and the sum of all 5 terms of the arithmetico-geometric progression is $\frac{49}{2}$, then a_4 is equal to _____ .

[JEE Mains-2023]

[Ans. : 16]

DIBY-59

If $7 = 5 + \frac{1}{7}(5 + \alpha) + \frac{1}{7^2}(5 + 2\alpha) + \frac{1}{7^3}(5 + 3\alpha) + \cdots \infty$, then the value of α is :

[JEE Mains-2025]

(A) $\frac{6}{7}$ (B) $\frac{1}{7}$ (C) 6 (D) 1

[Ans. : (C)]

A.G.P.

DIBY-60

Let $S = 2 + \frac{6}{7} + \frac{12}{7^2} + \frac{20}{7^3} + \frac{30}{7^4} + \cdots$. Then $4S$ is equal to [JEE Mains-2022]

(A) $\left(\frac{7}{3}\right)^2$ (B) $\frac{7^3}{3^2}$ (C) $\left(\frac{7}{3}\right)^3$ (D) $\frac{7^2}{3^3}$

[Ans. : (C)]

DIBY-61

If $\frac{6}{3^{12}} + \frac{10}{3^{11}} + \frac{20}{3^{10}} + \frac{40}{3^9} + \cdots + \frac{10240}{3} = 2^n \cdot m$, where m is odd, then mn is equal to [JEE Mains-2022]

[Ans. : 12]

DIBY-62

If $S = \frac{7}{5} + \frac{9}{5^2} + \frac{13}{5^3} + \frac{19}{5^4} + \cdots$, then $160S$ is equal to [JEE Mains-2021]

[Ans. : 305]

$S = r + 2r^2 + 3r^3 + \ldots\ldots + 60r^{60}$, $R = (1+r)$

DIBY-63

If $S(x) = (1+x) + 2(1+x)^2 + 3(1+x)^3 + \cdots + 60(1+x)^{60}, x \neq 0$, and $(60)^2 S(60) = a(b)^b + b$, where $a, b \in N$, then $(a+b)$ equal to [JEE Mains-2024]

[Ans. : 3660]

PROPERTIES OF SIGMA

DIBY-71

The value of $1^2 + 3^2 + 5^2 + \cdots + 25^2$ is : [JEE Mains-2013]

(A) 1728 (B) 1456 (C) 2925 (D) 1469

[Ans. : (C)]

DIBY-72

The sum of the series $1 + 2 \times 3 + 3 \times 5 + 4 \times 7 + \cdots$ upto 11th term is :

[JEE Mains-2019]

(A) 916 (B) 945 (C) 915 (D) 946

[Ans. : (D)]

DIBY-73

Find the sum of the first n terms of the series $3^2 + 7^2 + 11^2 + \cdots$ up to n terms

(A) $\frac{n}{3}(16n^2 + 12n - 1)$ (B) $\frac{n}{3}(16n^2 - 12n + 1)$

(C) $\frac{n}{3}(16n^2 + 12n + 1)$ (D) $\frac{n}{2}(16n^2 + 12n - 1)$

[Ans. : (A)]

PROPERTIES OF SIGMA

DIBY-74

Compute the sum of the series whose n^{th} term is given by $T_n = n(n+1)(3n-1)$.

(A) $\frac{n(n+1)(3n^2+3n-2)}{12}$

(B) $\frac{n(n+1)(9n^2+15n+2)}{12}$

(C) $\frac{n(n+1)(n+2)(9n-1)}{12}$

(D) $\frac{n(n+1)(3n^2+n+2)}{4}$

[Ans. : (C)]

DIBY-75

Compute the sum $(31)^2 + (32)^2 + (33)^2 + \cdots \ldots + (50)^2$.

(A) 32,925 (B) 33,470 (C) 34,185 (D) 35,310

[Ans. : (B)]

DIBY-76

The value(S) of the positive integer n for which the quadratic equation, $\sum_{k=1}^{n} (x+k-1)(x+k) = 10n$ has solutions α and $\alpha+1$ for some α is.

(A) 10 (B) 11 (C) 12 (D) 13

[Ans. : (B)]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module:

It is never about end result

It's always about Journey

Thank You

#futureIITians